TUM Master Practical Project Proposal 2026

Privacy Shield

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Git Repository: Privacy Shield

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1 Description

Privacy Shield is an IoT hardware ecosystem designed to protect acoustic privacy in indoor environments. The system consists of autonomous nodes attached to doors, windows, and other resonant surfaces. Each node captures ambient sound via an I2S MEMS microphone, detects human speech using Voice Activity Detection (VAD), and uses Audio Echo Cancellation (AEC) to remove feedback. It then emits an optimized masking sound, either pink or white noise, through a surface-mounted exciter transducer. The nodes are connected to a local web service, where users can monitor and calibrate them as needed. For offline users, an LCD touchscreen that functions as a tablet will connect the user to the service.

2 Background

Related work includes commercial acoustic masking systems such as the DNG-2300 generator [1], which has been created to protect against listening devices that cannot be discovered by common methods. The unit protects a room by inducting non-filterable noise onto surfaces. This noise, also known as "white noise", is transmitted onto surfaces with the help of the TRN-2000 transducers and OMS-2000 speakers in the unit.

Figure 1: DNG 2300 Illustration

3 Motivation

The main problem is straightforward: people outside a room can eavesdrop on conversations happening inside. At the same time, you don't want to disturb people outside the room with the masking noise you're generating to prevent that eavesdropping. Privacy Shield aims to solve both problems with an affordable, decentralized, DIY approach.

A single node costs around €50, and since the whole system runs on off-the-shelf components with open firmware, anyone can build it without special tools or professional installation. Existing commercial solutions, such as the DNG-2300, cost hundreds of euros and are designed for government or corporate use. Privacy Shield brings that same idea to a price point and form factor that actually makes sense for everyday spaces.

4 Architecture

The system employs a fully decentralized mesh topology. Each masking node is built around an ESP32-S3 microcontroller. The audio chain consists of an I2S MEMS microphone for capture and a Class-D amplifier driving a Dayton Audio transducer. Nodes communicate over ESP-NOW, a low-latency peer-to-peer protocol that requires no WiFi router or pairing. Each node makes independent masking decisions based on its own microphone input.

An adaptive masking algorithm scales the noise output proportionally to the detected speech energy. An adaptive filter handles acoustic echo cancellation to prevent feedback loops between the microphone and transducer.

An optional Hub (a standard ESP32-S3 and a tablet) can join the mesh over simultaneous WiFi and ESP-NOW. The Hub provides a Web-based dashboard accessible via any browser on the local network, offering per-node mute con-

trol, volume adjustment, battery status, and more.

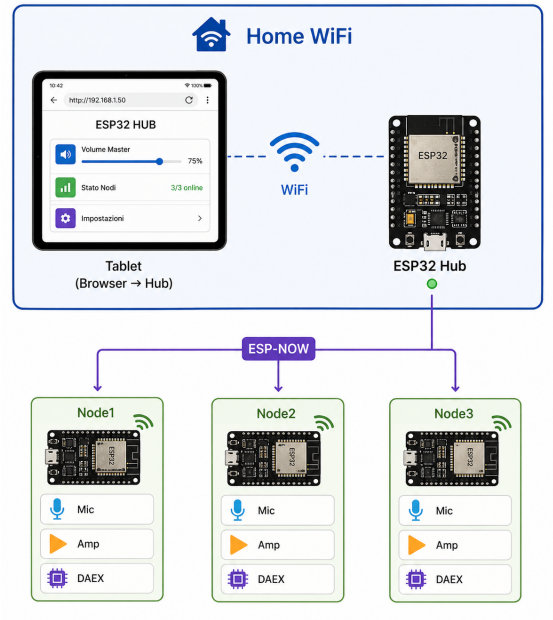

Figure 2: Network architecture

5 Project plan

The following section contains our plan to realize the project, dividing the workload over 2-week sprints from May to the end of July. Resulting in 4 sprints to complete the project.

5.1 Sprint 1

Set up ESP-IDF/FreeRTOS development environment. Implement the I2S microphone driver for raw audio capture. Use amplifiers and transducers to play tones and noise. Establish ESP-NOW basic communication with node discovery. Test mechanical isolation to minimize transducer vibration reaching the microphone.

5.2 Sprint 2

Implement double-buffered I2S capture. Develop energy-based and spectral Voice Activity Detection. Generate white and pink noise profiles. Implement the adaptive masking algorithm that scales output proportionally to input speech energy. Achieve fully autonomous per-node operation. Conduct real-room testing and parameter tuning.

5.3 Sprint 3

Implement an adaptive filter for echo path estimation. Integrate the full masking pipeline. Perform long-duration stability testing.

5.4 Sprint 4

Develop Hub firmware with simultaneous WiFi and ESP-NOW. Implement REST API endpoints. Build a Web dashboard with mute, volume, battery controls, and more features. Design and 3D-print enclosure with separate chambers for microphone and transducer isolation. Final system assembly and long-duration validation test.

6 Declaration

In preparing this document, we used artificial intelligence tools such as Gemini, DeepSeek, ChatGPT, and Grammarly to improve language, grammar, brainstorm structure, and to generate images. We critically reviewed, edited, and took full and sole responsibility for all content, results, and the final submitted version of this work.

A Hardware List

- ESP32-S3 8MB PSRAM Microcontroller (3x)
- I2S MEMS Microphone (3x)
- Dayton Audio DAEX25 Sound Exciter
- MAX98357A Class-D Amplifier
- 18650 Battery (4x)
- 18650 Battery Shield V3
- Breadboard (3x)
- USB connector (3x)
- Connectors
- 22 AWG Speaker Wire
- Tablet (LCD Touchscreen)

References

[1] DigiScan Labs. *DNG-2300*. https://digiscan-labs.com/products/dng-2300/. 2026.